

# Qwen3.6 Benchmark Report

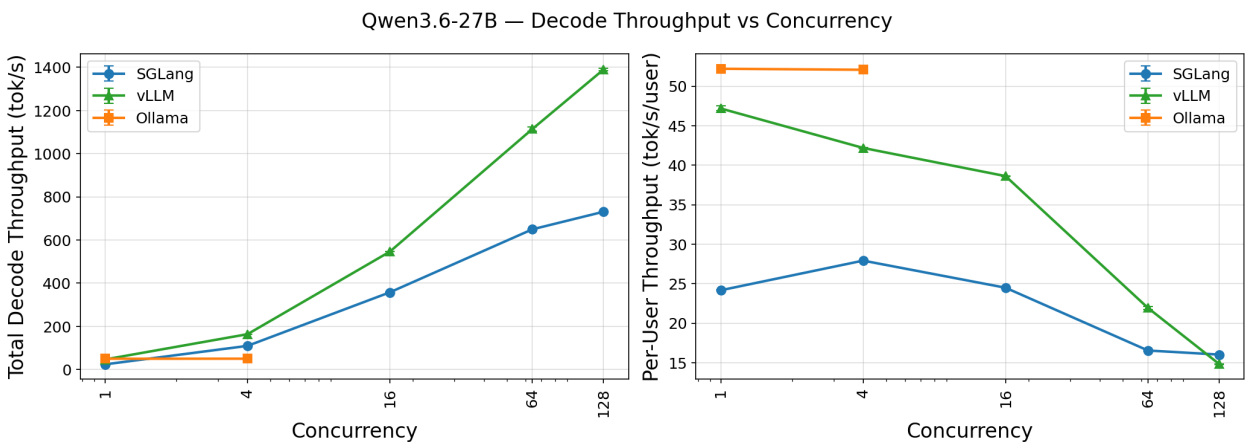
Inference throughput and latency comparison: SGLang vs vLLM vs Ollama  
Generated: 2026-05-03

## Section 1 — Qwen3.6-27B (SGLang vs vLLM vs Ollama)

### 1.1 Decode Throughput

Total decode throughput (tok/s) and per-user throughput (tok/s/user) across concurrency levels. Values are mean  $\pm$  1 std-dev over 3 runs.

| Concurrency | SGLang<br>Total (tok/s) | vLLM<br>Total (tok/s) | Ollama<br>Total (tok/s) | SGLang<br>Per-User (tok/s) | vLLM<br>Per-User (tok/s) | Ollama<br>Per-User (tok/s) |
|-------------|-------------------------|-----------------------|-------------------------|----------------------------|--------------------------|----------------------------|
| 1           | 24.1 $\pm$ 0.1          | 46.9 $\pm$ 0.3        | 50.1 $\pm$ 0.0          | 24.2 $\pm$ 0.1             | 47.2 $\pm$ 0.3           | 52.2 $\pm$ 0.0             |
| 4           | 109.7 $\pm$ 0.1         | 163.9 $\pm$ 0.2       | 50.3 $\pm$ 0.1          | 27.9 $\pm$ 0.0             | 42.2 $\pm$ 0.1           | 52.1 $\pm$ 0.1             |
| 16          | 357.9 $\pm$ 0.2         | 546.5 $\pm$ 0.4       | N/A                     | 24.5 $\pm$ 0.0             | 38.6 $\pm$ 0.1           | N/A                        |
| 64          | 649.6 $\pm$ 1.3         | 1114.1 $\pm$ 9.3      | N/A                     | 16.5 $\pm$ 0.1             | 21.9 $\pm$ 0.2           | N/A                        |
| 128         | 731.1 $\pm$ 0.2         | 1389.9 $\pm$ 5.1      | N/A                     | 16.0 $\pm$ 0.0             | 14.8 $\pm$ 0.0           | N/A                        |

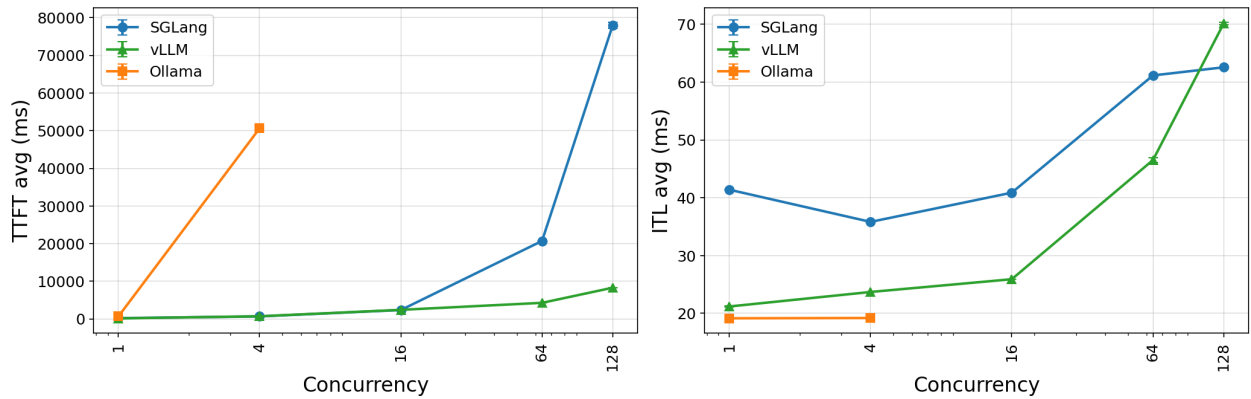


### 1.2 Decode Latency

Time-to-first-token (TTFT) and inter-token latency (ITL) statistics (avg  $\pm$  std-dev, p50) across concurrency levels.

| Concurrency | SGLang<br>TTFT<br>avg±std<br>(ms) | vLLM<br>TTFT<br>avg±std<br>(ms) | Ollama<br>TTFT<br>avg±std<br>(ms) | SGLang<br>TTFT<br>p50<br>(ms) | vLLM<br>TTFT<br>p50<br>(ms) | Ollama<br>TTFT<br>p50<br>(ms) | SGLang<br>ITL<br>avg±std<br>(ms) | vLLM<br>ITL<br>avg±std<br>(ms) | Ollama<br>ITL<br>avg±std<br>(ms) | SGLang<br>ITL p50<br>(ms) | vLLM<br>ITL p50<br>(ms) | Ollama<br>ITL p50<br>(ms) |
|-------------|-----------------------------------|---------------------------------|-----------------------------------|-------------------------------|-----------------------------|-------------------------------|----------------------------------|--------------------------------|----------------------------------|---------------------------|-------------------------|---------------------------|
| 1           | 224.5 ± 2.5                       | 163.8 ± 1.3                     | 686.6 ± 0.6                       | 223.0                         | 163.8                       | 686.5                         | 41.4 ± 0.1                       | 21.2 ± 0.1                     | 19.2 ± 0.0                       | 41.4                      | 21.2                    | 19.2                      |
| 4           | 688.2 ± 0.4                       | 723.3 ± 35.2                    | 50638.3 ± 614.0                   | 843.0                         | 811.3                       | 52248.0                       | 35.8 ± 0.0                       | 23.7 ± 0.0                     | 19.2 ± 0.0                       | 35.8                      | 23.7                    | 19.2                      |
| 16          | 2380.7 ± 48.2                     | 2429.1 ± 35.8                   | N/A                               | 1905.0                        | 2342.4                      | N/A                           | 40.9 ± 0.0                       | 25.9 ± 0.0                     | N/A                              | 41.1                      | 25.9                    | N/A                       |
| 64          | 20715.1 ± 87.9                    | 4288.7 ± 28.0                   | N/A                               | 8411.4                        | 3720.2                      | N/A                           | 61.2 ± 0.2                       | 46.5 ± 0.4                     | N/A                              | 61.8                      | 46.5                    | N/A                       |
| 128         | 77916.4 ± 811.4                   | 8302.3 ± 74.2                   | N/A                               | 79646.9                       | 5434.5                      | N/A                           | 62.6 ± 0.0                       | 70.1 ± 0.2                     | N/A                              | 62.4                      | 70.1                    | N/A                       |

Qwen3.6-27B — Decode Latency vs Concurrency

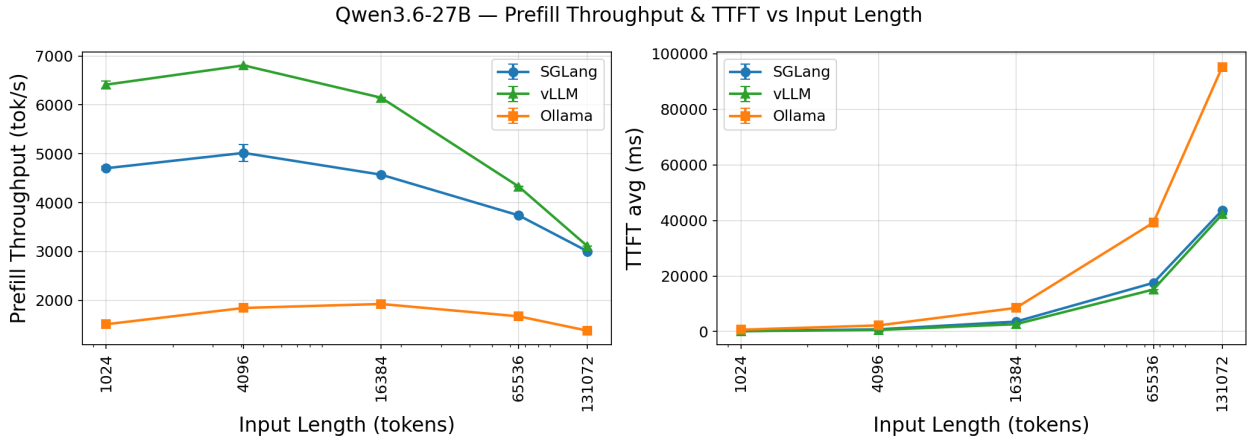


### 1.3 Prefill Performance

Prefill throughput (computed as  $\text{input\_len} / \text{TTFT}$ ) and TTFT statistics across input sequence lengths. Error propagation:  $\sigma_{\text{tput}} = \text{tput} \times (\sigma_{\text{TTFT}} / \text{TTFT\_avg})$ .

| Input Len | SGLang<br>Tput<br>avg±std<br>(tok/s) | vLLM<br>Tput<br>avg±std<br>(tok/s) | Ollama<br>Tput<br>avg±std<br>(tok/s) | SGLang<br>TTFT<br>avg±std<br>(ms) | vLLM<br>TTFT<br>avg±std<br>(ms) | Ollama<br>TTFT<br>avg±std<br>(ms) | SGLang<br>TTFT<br>p50<br>(ms) | vLLM<br>TTFT<br>p50<br>(ms) | Ollama<br>TTFT<br>p50<br>(ms) |
|-----------|--------------------------------------|------------------------------------|--------------------------------------|-----------------------------------|---------------------------------|-----------------------------------|-------------------------------|-----------------------------|-------------------------------|
| 1024      | 4702 ± 40                            | 6413 ± 73                          | 1509 ± 14                            | 217.8 ± 1.8                       | 159.7 ± 1.8                     | 678.7 ± 6.2                       | 217.8                         | 159.8                       | 678.7                         |
| 4096      | 5016 ± 172                           | 6807 ± 4                           | 1843 ± 1                             | 817.2 ± 27.5                      | 601.7 ± 0.4                     | 2222.6 ± 1.1                      | 833.8                         | 598.6                       | 2219.6                        |
| 16384     | 4570 ± 4                             | 6146 ± 3                           | 1923 ± 1                             | 3585.2 ± 3.2                      | 2666.0 ± 1.2                    | 8517.9 ± 3.7                      | 3592.6                        | 2664.2                      | 8516.2                        |
| 65536     | 3737 ± 2                             | 4327 ± 4                           | 1671 ± 3                             |                                   |                                 |                                   | 17536.5                       | 15146.1                     | 39215.9                       |

| Input Len | SGLang Tput avg±std (tok/s) | vLLM Tput avg±std (tok/s) | Ollama Tput avg±std (tok/s) | SGLang TTFT avg±std (ms) | vLLM TTFT avg±std (ms) | Ollama TTFT avg±std (ms) | SGLang TTFT p50 (ms) | vLLM TTFT p50 (ms) | Ollama TTFT p50 (ms) |
|-----------|-----------------------------|---------------------------|-----------------------------|--------------------------|------------------------|--------------------------|----------------------|--------------------|----------------------|
|           |                             |                           |                             | 17536.8 ± 7.8            | 15146.2 ± 12.8         | 39228.2 ± 68.0           |                      |                    |                      |
| 131072    | 3002 ± 1                    | 3106 ± 1                  | 1377 ± 10                   | 43654.8 ± 20.0           | 42203.1 ± 17.6         | 95168.8 ± 697.8          | 43651.8              | 42202.4            | 95010.7              |

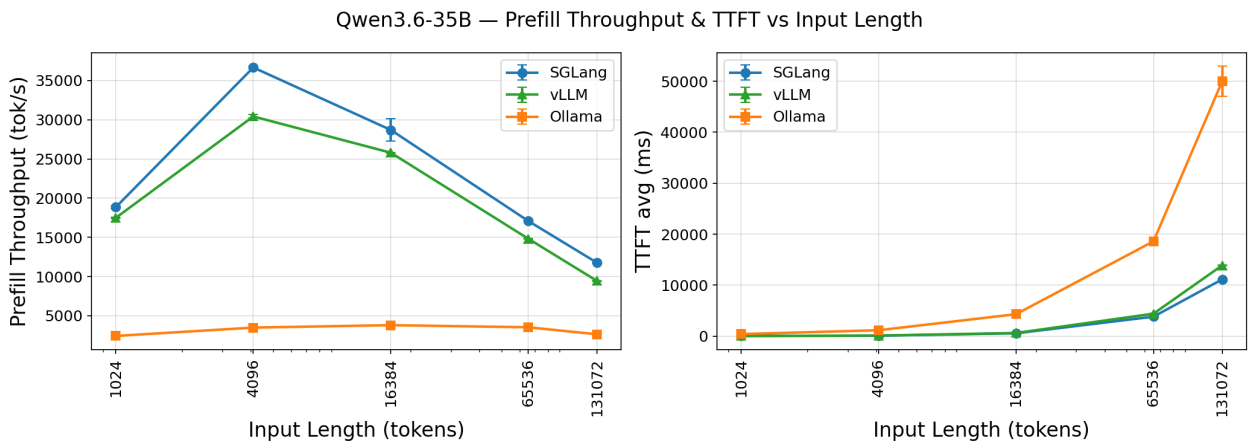


# Section 2 — Qwen3.6-35B (Available Engines)

## 2.1 Prefill Performance

Decode benchmarks for Qwen3.6-35B are pending. Prefill results are shown for available engines only.

| Input Len | SGLang Tput<br>avg±std<br>(tok/s) | vLLM Tput<br>avg±std<br>(tok/s) | Ollama Tput<br>avg±std<br>(tok/s) | SGLang TTFT<br>avg±std<br>(ms) | vLLM TTFT<br>avg±std<br>(ms) | Ollama TTFT<br>avg±std<br>(ms) | SGLang TTFT<br>p50<br>(ms) | vLLM TTFT<br>p50<br>(ms) | Ollama TTFT<br>p50<br>(ms) |
|-----------|-----------------------------------|---------------------------------|-----------------------------------|--------------------------------|------------------------------|--------------------------------|----------------------------|--------------------------|----------------------------|
| 1024      | 18837 ± 196                       | 17480 ± 64                      | 2414 ± 2                          | 54.4 ± 0.6                     | 58.6 ± 0.2                   | 424.2 ± 0.3                    | 54.2                       | 58.5                     | 423.7                      |
| 4096      | 36658 ± 165                       | 30414 ± 238                     | 3482 ± 4                          | 111.7 ± 0.5                    | 134.7 ± 1.1                  | 1176.3 ± 1.2                   | 111.8                      | 134.5                    | 1176.2                     |
| 16384     | 28694 ± 1417                      | 25789 ± 46                      | 3791 ± 1                          | 571.9 ± 27.5                   | 635.3 ± 1.1                  | 4321.9 ± 0.8                   | 631.5                      | 635.4                    | 4323.2                     |
| 65536     | 17079 ± 5                         | 14818 ± 21                      | 3522 ± 9                          | 3837.3 ± 1.1                   | 4422.7 ± 6.4                 | 18605.7 ± 47.7                 | 3835.2                     | 4421.6                   | 18598.8                    |
| 131072    | 11753 ± 48                        | 9431 ± 41                       | 2627 ± 155                        | 11152.4 ± 45.2                 | 13898.2 ± 60.9               | 49984.3 ± 2951.9               | 11149.3                    | 13872.7                  | 48268.2                    |



# Summary

## Qwen3.6-27B — Key Findings

- At concurrency 128, **vLLM** achieves the highest total decode throughput at **1390 tok/s**.
- At concurrency 1, **Ollama** has the lowest ITL at **19.2 ms**, indicating the fastest single-user decoding speed.
- For 128K-token prefill, **vLLM** leads with **3106 tok/s** prefill throughput.
- Ollama decode results are available for concurrency [1, 4], with concurrency [16, 64, 128] not yet benchmarked.

## Qwen3.6-35B — Key Findings

- At 1024-token input, **SGLang** achieves **18837 tok/s** prefill throughput for the 35B model.
- Ollama prefill throughput on 35B scales from 2414 tok/s (1K) to 2627 tok/s (128K), reflecting memory-bandwidth saturation at long contexts.
- The 35B model is expected to show lower throughput compared to 27B due to increased parameter count; full engine comparisons are forthcoming.